

Woodworking for Complete Beginners - Printable Sheets

From *Woodworking for Complete Beginners* by Arthur Kelsey
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Take This to the Shop: Nominal vs Actual

The board is not the size on the label. US softwood finished sizes:

Sold as	Actually measures (in)	Actually measures (mm)
1 x 2	3/4 x 1-1/2	19 x 38
1 x 4	3/4 x 3-1/2	19 x 89
1 x 6	3/4 x 5-1/2	19 x 140
1 x 8	3/4 x 7-1/4	19 x 184
1 x 10	3/4 x 9-1/4	19 x 235
1 x 12	3/4 x 11-1/4	19 x 286
2 x 2	1-1/2 x 1-1/2	38 x 38
2 x 4	1-1/2 x 3-1/2	38 x 89
2 x 6	1-1/2 x 5-1/2	38 x 140

Planed timber sold elsewhere (PAR / PSE) is also undersized, usually by 3-5 mm per dimension. **Plywood is often slightly under its stated thickness too.** Measure it.

Picking a Board - the 60-Second Check

	Checked
Sight along its length like aiming a rifle - reject bends you can see	

Cup - curved across the width	
Bow - curved along the length, face on	
Crook - curved along the length, edge on	
Twist - ends in different planes. Reject it	
Lay it on a flat floor - does it rock?	
Ends split? Knots loose enough to push with a thumb?	
Does it feel cold and heavy for its size? (too wet - it will move)	

Measuring and Marking - Crib Sheet

The rules	
One tape measure for the whole project	Tapes disagree with each other
Sharp pencil or a knife	A blunt line is a millimetre wide
Mark an X on the waste side, every time	Before you cut
Cut on the waste side, leaving the line just visible	You can take more off; you cannot put it back
Measure from one reference edge, always the same one	
Better still: do not measure - hold the piece where it goes and mark it there	

Is my square actually square? Hold it against a straight edge and draw a line. Flip it over, same edge, same point, draw again. The two lines should sit exactly on top of each other. A narrow V means the square is wrong - and that error goes into everything you build.

Is this assembly square? Measure both diagonals. Equal means square. Do it **before** the glue sets.

Project 1 - Bench Hook

Teaches: sawing to a line, a square corner, pilot holes. **Time:** 1 hour.

Stock: 1 x 8 x 3 ft; 1 x 2 x 2 ft. Plus four 1-1/4 in screws and glue.

Part	Qty	Size
Base	1	7-1/4 x 12 in
Stops	2	3/4 x 1-1/2 x 7-1/4 in

Order: cut base - cut stops - glue + screw one stop on top at one end - glue + screw the other underneath at the other end - ease edges.

Where people go wrong: stops not flush with the sides; no pilot hole, so the stop splits; both stops on the same face.

Project 2 - Cutting Board

Teaches: edge-to-edge glue-up, flattening, a food-safe finish.

Time: an evening + overnight.

Stock: hardwood 3/4 in thick (maple, beech, cherry, walnut). Not oak, ash, or any softwood.

Part	Qty	Size
Strips	5	3/4 x 1-1/2 x 12 in

Finished: 7-1/2 x 12 in.

Order: cut 5 strips slightly long - arrange and mark a triangle across them - dry-clamp - glue up, 3 clamps - overnight - pare off squeeze-out - sand 80/120/180/220 - raise the grain twice - ease all edges - mineral oil daily for 3-4 days.

Where people go wrong: too few clamps; squeeze-out left on, so the oil blotches; cooking oil instead of mineral oil (it goes rancid); dishwasher.

Project 3 - Shelf

Teaches: drilling, countersinking, fixing to a wall. **Time:** 2 hours.

Stock: 1 x 8 x 3 ft - that is all; brackets come from the offcut.

Part	Qty	Size
Shelf	1	7-1/4 x 24 in
Bracket blanks	2	6 x 6 in, each cut corner to corner

36 in board – 24 in shelf = 12 in = exactly two 6 in blanks. A 6 in square will **not** come out of a 1 x 4 (only 3-1/2 in wide).

Order: cut shelf - cut two 6 in squares, cut each diagonally - find the studs **first** - mark level - fix brackets to wall - sit shelf on, check level, screw down (pilot + countersink).

Where people go wrong: ordinary screws into plasterboard - it will fail weeks later with books on it; brackets not level with each other.

Project 4 - Box

Teaches: rebate joints, four corners that meet, checking square.

Time: an afternoon + glue.

Stock: 1 x 4 x 4 ft; 1/4 in plywood.

Part	Qty	Size
Long sides	2	3/4 x 3-1/2 x 10 in, rebate both ends 3/4 wide x 3/8 deep
Short sides	2	3/4 x 3-1/2 x 6 in
Bottom	1	6-3/4 x 10 in, 1/4 in ply

Finished: 10 x 6-3/4 in (6 in + 3/8 in lip at each end).

Order: cut sides in matched pairs, clamped together - mark rebates on inside faces - saw shoulder, pare waste - dry-fit and check diagonals - glue, clamp, re-check diagonals - when dry, stand the box on the ply and draw round it - glue and pin the bottom.

Where people go wrong: rebate too deep so the lip breaks; not checking diagonals before it sets; forcing a tight joint and splitting the lip.

Project 5 - Step Stool

Teaches: taking weight, bracing. **Time:** an afternoon.

Stock: 1 x 10 x 4 ft; 1 x 4 x 2 ft. Plus twelve 1-1/4 in screws.

Part	Qty	Size
Top	1	9-1/4 x 14 in
Sides	2	9-1/4 x 9 in
Brace	1	3-1/2 x 12-1/2 in

Brace = $14 - (2 \times 3/4) = 12-1/2$ in. Finished height 9-3/4 in.

Order: cut sides together, clamped, so they match - cut top - screw top to sides (4 screws each, pilot + countersink) - fit brace between sides, 2 in off the floor - check for rock - round the top edges well - test your weight gradually at each corner.

Where people go wrong: leaving out the brace (it racks sideways under someone standing on the edge); sides cut to different heights; screws too near the end, splitting the side.

Project 6 - Side Table

Teaches: legs and frame, a glued top, wood movement. **Time:** a weekend.

Stock: three 1 x 6 x 2 ft; 1 x 4 x 5 ft; 2 x 2 x 6 ft.

Part	Qty	Size
Top strips	3	3/4 x 5-1/2 x 15 in
Front/back aprons	2	3/4 x 3-1/2 x 15 in
Side aprons	2	3/4 x 3-1/2 x 12 in
Legs	4	1-1/2 x 1-1/2 x 17-1/4 in

Top 16-1/2 x 15 in. Frame 15 x 13-1/2 in. Overhang 3/4 in all round. Height 18 in.

Order: glue up the top, leave overnight - cut all four legs together - build the apron frame, check diagonals - fit legs into the inside corners, screwed through both aprons - check it does not rock - **attach the top with turn buttons or slotted fasteners, never screwed down solid.**

Where people go wrong: screwing the top down rigidly - a 16-1/2 in solid panel moves across its width and will split; legs of unequal length; finishing only the top face, so the panel cups.

Finishing Schedule

Step	
1	Sand to 180 (or 220 for oil). Check in raking light
2	Vacuum everything, including the room. Let the dust settle
3	Wipe on a generous coat
4	Let it soak 15-20 minutes
5	Wipe off ALL the excess, thoroughly - the step everyone skips, and the reason finishes go sticky
6	Leave 12-24 hours
7	Light rub with 320 grit or a grey pad
8	Second coat. Two or three total is plenty

Finish both faces of any wide panel, or it will cup.

Fire warning - read this one

Rags soaked in linseed, tung or Danish oil can catch fire on their own. The oil generates heat as it cures; a crumpled rag insulates that heat and can reach ignition with nothing else involved. It is a documented cause of house fires.

Do this, every time	
Spread rags flat and singly, outdoors, weighted, until completely stiff and dry	
OR drop them in a sealed metal	

container of water	
Never ball them up. Never leave them in a bin	